ZCAS UNIVERSITY

SCHOOL OF COMPUTING, TECHNOLOGY & APPLIED SCIENCES

CAS5821 MATHEMATICAL FOUNDATIONS OF DATA SCIENCE

FINAL EXAMINATION

DATE

TIME

TIME ALLOWED : WRITING – THREE HOURS

READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: This question is **compulsory** and must be attempted.
2. Sections B: Answer **TWO (2)** questions from this section.
3. This examination paper carries a total of **100 marks.**
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted.

Question 1

(a) A store is doing inventory for one week to determine the popularity of a new product

Day Number	1	2	3	4	5	6	7
Number sold MB	24	32	41	50	60	68	77

i) Find the *linear regression equation* that models this data. (Round to *nearest integer.)*
Choose:

$y = 8x + 14$ $y = 9x + 15$

$y = 15x + 9$ $y = 14x + 8$ [15 Marks]

ii) If this pattern of growth continues, predict the number sold on the 10^{th}day.
Choose:

94 105 148 159

[15 Marks]

(b) Let $A = \begin{pmatrix} 1 & -1 & -2 \\ 0 & 2 & -1 \\ -1 & 3 & 1 \end{pmatrix}$. Find all the Eigen Values and Eigen Vectors of A.

[20 Marks]

(50 marks)

SECTION B: Attempt any TWO (2) questions in this section

Question 2

Consider the summary output below:

SUMMARY
OUTPUT

Regression Statistics	
Multiple R	0.67565694
R Square	0.45651229
Adjusted R Square	0.35769635
Standard Error	0.48959527
Observations	66

ANOVA

	df	SS	MS	F	Significance F
Regression	10	11.07388	1.107388	4.619824	9.07E-05
Residual	55	13.18369	0.239704		
Total	65	24.25758			

a) What is the meaning of the coefficient of determination (R Square) above?

[10 Marks]

b) Draw conclusions for the above output to determine whether the independent variables have an effect on the dependent variable. [15 Marks]

(25 marks)

Question 3

a) Give five examples where regression Analysis can be used in solving real life problems.

[10 Marks]

b) Calculate the product moment correlation coefficient and hence the coefficient of determination (r^2) for the set of data given below and interpret the results:

X	3	4	6	7	8	10	12	15	16
Y	3.2	2.9	3.7	2.2	1.8	2.3	1.7	0.8	0.3

[15 Marks]

(Total: 25 marks)

Question 4

a) An expert backgammon player is currently competing in two tournaments. He estimates the probability of winning the first tournament as 0.6 and the probability of winning the second tournament as 0.5. If he wins the first tournament, he estimates the probability of winning the second tournament increases to 0.65 because of an increase in confidence. What is the probability of:

(i) the player winning both tournaments? [8 Marks]

(ii) the player winning at least one tournament? [7 Marks]

b) Find $\frac{dy}{dx}$ of the following:

(i) $y = 4x^3 + 5x^2 - 1$ (ii) $y = \sqrt{2x-1}$ [10 Marks]

(25 marks)

END OF EXAMINATION

$\int 12x^{2+1}$

$= \frac{12x^3}{3}$

$4x^3$